

Mth 202-Mid by Mm



Q.1

Determine whether the function $f(x) = x+1$ is one-to-one.

Ans:-

Let

$$f(x_1) = f(x_2) \quad (x_1, x_2 \in \mathbb{R})$$

$$x_1 + 1 = x_2 + 1 \quad (\text{by the definition of } f)$$

Subtract 1 both side

$$x_1 + 1 - 1 = x_2 + 1 - 1$$

$$x_1 = x_2$$

~~Mm~~

Hence $f(x)$ is one-to-one.

Q.2

Determine whether the function

$$x_1 = x_2$$

Hence $f(x)$ is one-to-one.

Q.2

Determine whether the function
 $f(x) = x^3 + 5$ is onto.

Ans:-

let

$$y = f(x)$$

$$y = x^3 + 5 \rightarrow (i) \text{ (by definition of } f)$$

$$y - 5 = x^3$$

$$(y - 5)^{1/3} = (x^3)^{1/3}$$

$$\sqrt[3]{y - 5} = x$$

put x in (i)

$$f(x) = (\sqrt[3]{y - 5})^3 + 5$$

$$= y - 5 + 5$$

$$= y$$

Hence $f(x)$ is onto

Mth202-Mid

Q.3

Compute the first four terms of the sequence defined by

$$a_n = 3n - 5$$

Ans:-

M.M.

Given, $a_n = 3n - 5 \rightarrow (i)$

Now we find first four terms.

Put $n = 1, 2, 3, 4$ in (i)

When $n = 1$, $a_1 = 3(1) - 5 = -2$

When $n = 2$, $a_2 = 3(2) - 5 = 1$

When $n = 3$, $a_3 = 3(3) - 5 = 4$

When $n = 4$, $a_4 = 3(4) - 5 = 7$

Hence first four terms are

$-2, 1, 4, 7$

M.M.

Q.4

show that $P(x) = x^3 + 1$ is onto

Hence first four

-2, 1, 4, 7

M.M 3

Q.4

Show that $f(x) = x^3 + 1$ is onto function.

Ans: - let $y = f(x)$

$$y = x^3 + 1 \rightarrow (i)$$

$$y - 1 = x^3$$

$$\sqrt[3]{y-1} = x$$

put the value of

x in (i)

~~$y =$~~ $\hookrightarrow$

$$f(x) = y$$

$$y = \left(\sqrt[3]{y-1}\right)^3 + 1$$

$$y = y - 1 + 1$$

$$y = f(x)$$

Hence $f(x)$ is onto.

Mth 202-Mid

Q.5

Find the sum of the infinite
G.P. $2, \sqrt{2}, 1, \dots$

Ans:-

~~Given~~ Given

$2, \sqrt{2}, 1, \dots$

M.M

$a = \text{First term} = 2$

$r = \text{Common ratio} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$

$\therefore |r| < 1$

$$\text{Sum} = S = \frac{a}{1-r} = \frac{2}{1 - \frac{1}{\sqrt{2}}}$$

$$= \frac{2}{\frac{\sqrt{2}-1}{\sqrt{2}}}$$

$$= 2 \div \frac{\sqrt{2}-1}{\sqrt{2}}$$

$$= 2 \times \frac{\sqrt{2}}{\sqrt{2}-1} = \frac{2\sqrt{2}}{\sqrt{2}-1}$$

Q.6

Define Finite and Infinite Set.

$$= 2 \div \frac{1}{\sqrt{2}}$$

$$= 2 \times \frac{\sqrt{2}}{\sqrt{2}-1} = \frac{2\sqrt{2}}{\sqrt{2}-1}$$

Q.6

Define Finite and Infinite set.

Ans:-

Finite set:-

~~Nm~~

A set is called finite if and only if it is empty set or there is one-to-one correspondence from $\{1, 2, 3, \dots, n\}$ to this set. **Infinite set.** A non empty set that cannot be put into one-to-one correspondence with $\{1, 2, 3, \dots, n\}$ to this set.

Every integer $2n$ is the image of an integer n .

Q.7

Let $f(x) = ax + b$ and $g(x) = cx + d$ where a, b, c and d are constant. Under what condition $f \circ g(x) = g \circ f(x)$.

Ans: -

mm

Given, $f(x) = ax + b$, $g(x) = cx + d$

$$f \circ g(x) = f(g(x)) = f(cx + d) = a(cx + d) + b \\ = acx + ad + b$$

$$g \circ f(x) = g(f(x)) = g(ax + b) = c(ax + b) + d \\ = acx + bc + d$$

$$acx + ad + b = acx + bc + d$$

$$ad + b = bc + d \quad \text{Hence } f \circ g(x) = g \circ f(x)$$

Mth202-Mid

Q.8

Show that the set $2\mathbb{Z}$ of all even integers is countable.

Ans:-

Define $h: \mathbb{Z} \rightarrow 2\mathbb{Z}$ by

$$h(n) = 2n \quad \forall n \in \mathbb{Z}$$

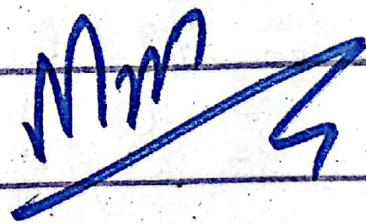
one-to-one

$$h(n_1) = h(n_2) \quad \text{then}$$

$$2n_1 = 2n_2$$

$$n_1 = n_2$$

onto



Every integer $2n$ is the image of an integer n .

Q.7

let $f(x) = ax + b$ and $g(x) = cx + d$ where

Q9

let $A = \{a, b, c, d\}$ be a set and consider the relation $R = \{(a, a), (a, b), (a, c), (a, d), (b, d), (b, b), (c, c), (c, d), (d, d)\}$ on A . Show that R is a partial ordering.

Ans:-

R is partial ordering on set A if it is reflexive, transitive and antisymmetric.

Reflexive.

M.M

R is reflexive because $(a, a), (b, b), (c, c)$ and (d, d) , these pairs are the elements of A .

Antisymmetry.

R is antisymmetry because there is no such pairs i.e. (x, y) and (y, x) it means $x = y$.

M.M

Transitivity.

R is transitive because for any three elements a, b and c or b, c and d in set A . it means (a, b) and (a, c) or (b, d) and (c, d) are pairs of R .

Hence R is partial ordering.

Mth 202 - Mid

Q. 10

If $R = \{(1, y), (2, x), (2, y), (3, x)\}$

Find domain and range of the relation R.

Ans. -

$$\text{Domain} = \{1, 2, 3\}$$

$$\text{Range} = \{x, y\}$$

M.M

Note: - دو بار سے زیادہ بار ایک سے آئے والے
عنصر کو ایک بار لکھنا ہے Range میں بھی
اور domain میں بھی

Q. 11

Note: - ہر ایک بار گنا ہے Range میں ہے۔
 اور domain میں ہے۔

Q.11

Show that $p \rightarrow q \equiv \neg q \rightarrow \neg p$

Ans:-

p	q	$\neg q$	$\neg p$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

M.M

Hence

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

Mth202-Mid

Q.12

Show that $q \rightarrow p \equiv \neg p \rightarrow \neg q$

Ans:-

p	q	$q \rightarrow p$	$\neg p$	$\neg q$	$\neg p \rightarrow \neg q$
T	T	T	F	F	T
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	T	T

Hence

$$q \rightarrow p \equiv \neg p \rightarrow \neg q$$

N.M